Summary Report: Candidate literary paper to be used

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Contents

1 Paper 1: Optimum government size and economic growth in case of Indian states: Evidence from panel threshold model

1.1 Summary

This paper investigates the relationship between optimum government size and economic growth using data of Indian states during 1990-91 to 2017–18. Our results derived from panel threshold regression model show a positive and significant impact of government size on economic growth within the estimated thresholds for both aggregate and sub-panels based on income and regions. Once the government size moves above the upper threshold level, then its impact declines and turns to be insignificant. Thus, our findings suggest the policymakers for maintaining the government size within the thresholds limit. Akram and Rath 2020

1.2 Data sources

- Indian states during 1990-91 to 2017–18
- Gross State Domestic Products (GSDP, 2011–12) and expenditures (such total expenditure, social sector expenditure
- revenue expenditure, and capital expenditure) data collected from Economic and Political Weekly Research Foundation (EPWRF).

1.3 Methodology

theoretical model which is proposed by Barro (1990, 1991).

$$Y(t) = AK(t)^{\beta}G(t)^{1-\beta}, \quad 0 < \alpha < 1$$

The thresholds model is as follows:

$$\delta y_{i,t} = \alpha_{)} + \pi_i t + \beta_i y_{t-1} + \sum_{j=1}^{k} \psi_{i,j} \delta y_{i,t-1} + \epsilon_{i,t}$$

1.4 Conclusions

- **Objective:** The study addresses the literature gap regarding sub-national economic growth by examining the relationship between optimum government size and growth across major Indian states (1990–91 to 2017–18).
- **Methodology:** A panel threshold regression model was employed, identifying two significant thresholds for the aggregate panel and most sub-panels.
- **Optimal Thresholds (Aggregate):** Government size has a positive and significant effect on economic growth specifically when it lies between the lower and upper thresholds of 4.85% and 5.09%.
- **Category Variations:**
  - **NSC States:** Estimated optimal threshold range is 4.55%–6.07%.
  - **SC States:** Estimated optimal threshold range is significantly higher at 34.70%–38.55%.
- **Diminishing Returns:** Once government size exceeds the upper threshold, the coefficient of growth declines and becomes statistically insignificant.
- **Expenditure Composition:** Findings remain consistent when disaggregated into revenue, capital, and social sector expenditures, showing positive growth associations within threshold limits.
- **Policy Recommendations:** State governments should utilize fiscal rules and budgetary discipline to maintain an ideal size, ensuring that Keynesian deficit spending remains within the "optimum" range to avoid unfavorable growth consequences.

2 Paper 2: Does too much government spending depress the economic development of transition economies? Evidences from dynamic panel threshold analysis

2.1 Summary

This paper investigates the relationship between government size and economic growth and determines the optimal level of government spending to maximize economic growth. The paper applies a dynamic panel data analysis based upon a threshold model to test the threshold effect of government spending in 26 transition economies over the period spanning 1993–2016. According to the analysis results, government expenditures have a threshold effect on economic growth, and there is a non-linear relationship depicted as an Armey curve in these transition economies. The findings indicate that a government size above the threshold government spending level adversely affects economic growth, while a government size below the threshold level has a positive effect. Furthermore, there is a statistically significant relationship between the two variables above and below that optimal level, even if we divide the sample into developed and developing countries. Our findings suggest that governments in transition economies should consider optimal government size at around the estimated threshold level to support sustainable economic growth.Aydin and Esen 2019

2.2 Methodology

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$$Y_{it} = \alpha_0 + \alpha_i initial_{it} + \alpha_2 Gov_{it} + \beta x_{it} + \epsilon_{it}$$

Where:

- Y_{it}: Real GDP per capita growth rate in country i at time t.
- α_0: Constant term.
- $initial_{it}$: Initial level of income in country i at time t.
- α_1: Coefficient measuring the effect of initial income on economic growth.
- Gov_{it}: Government spending level in country i at time t.
- α_2: Coefficient capturing the impact of government spending on economic growth.
- x_{it}: Vector of other macroeconomic control variables affecting economic growth.
- β: Vector of coefficients associated with the control variables.
- ε_{it}: White noise error term.

2.3 Conclusions

- The relationship between government expenditures and economic growth is widely debated, but has rarely been studied in the context of transition economies moving from centrally planned to free market systems.
- This study empirically examines threshold effects of government size on economic growth in 25 transition economies from 1993 to 2016.
- Using a threshold autoregressive (TAR) approach, the study avoids assuming a specific functional form and identifies optimal government size more accurately.
- Results show a non-linear relationship: government expenditures positively affect growth below a threshold, but this effect weakens and becomes negative (though insignificant) beyond it.
- The estimated threshold levels are 11.67 % of GDP for developing economies and 17.54 % for developed economies.
- Government intervention supports growth through education, health, inequality reduction, and institutional development, but excessive intervention leads to diminishing returns.
- Beyond the optimal size, government expansion may hinder growth due to taxation burdens, inefficiencies, corruption, rent-seeking, and crowding out of private investment.
- Transition economies require some degree of government intervention (e.g., liberalization, privatization, legal reforms) to support market functionality during the transition process.
- Sustainable growth depends on effective and limited government intervention; exceeding the optimal size does more harm than good.
- The study also highlights that government impact extends beyond expenditure size to include regulatory, legal, and bureaucratic interventions.

3 Paper 3: GOVERNMENT SIZE AND ECONOMIC GROWTH IN TURKEY: A THRESHOLD REGRESSION ANALYSIS

3.1 Summary

We examine the relationship between the government size and economic growth by using threshold regression model and quarterly data over the period 1998:1–2015:1 for Turkey. Our results provide a strong evidence for the existence of a non-linear relationship. The estimated threshold levels, as a percentage of GDP, are 16.5 for the government total expenditures, 12.6 for consumption expenditures and 3.9 for investment expenditures. We find that an increase in the government size leads to a significant rise (decline) in economic growth rate when the government size is below (above) the threshold level, confirming the predictions of Armey curve. Our findings have a clear policy implication: since the realized government consumption and total expenditures are well above the estimated threshold levels, a reduction in the government size would boost the growth rate. Iyidogan and Turan 2017

3.2 Methodology

$$y_i = \alpha_0 + alpah_1(\frac{I_t}{Y_t}) + \alpha_2 Gov_t + u_t$$

- Y: GDP growth rate.
- C/Y: Government consumption expenditure (% of GDP).
- GI/Y: Government sector gross fixed capital formation (% of GDP).
- G/Y: Total government expenditure (% of GDP).
- GOV: The product of government expenditure growth and government size (government expenditure / GDP).
- I/Y: Private sector gross fixed capital formation (% of GDP).

3.3 Conclusions

- **Non-linear Relationship:** The study identifies a non-linear, non-monotonic relationship between government size and economic growth in Turkey (1998:1–2015:1), confirming the validity of the *Armey curve*.
- **Threshold Estimates:** Threshold levels as a share of GDP are estimated at 16.5% for total expenditures (G/Y), 12.6% for consumption expenditures (C/Y), and 3.9% for investment expenditures (GI/Y).
- **Growth Dynamics:** Government expansion significantly boosts economic growth when below the threshold level, but has a detrimental effect once the size exceeds these optimal limits.
- **Current Fiscal Status:** Realized total and consumption expenditures in Turkey have consistently remained above the estimated thresholds, while investment expenditures exceeded the threshold following the 2003–2009 stabilization period.
- **Policy Implications:** Findings suggest that to boost growth, policy makers should prioritize spending efficiency and restrain unproductive expenditures, particularly given future fiscal pressures such as an ageing population.
- **Future Fiscal Risk:** Projected increases in health and social security spending due to demographic shifts may further expand government size beyond optimal levels if structural reforms are not implemented.

References

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